Whale-Watching Tourism vs. the Disturbance Behind It

1. The Issue

Whale watching is one of the fastest-growing forms of marine tourism worldwide. It allows people to encounter animals they may otherwise never see, but the closeness that makes the experience appealing also creates the problem: the vessels that make it possible can disturb whales through underwater noise, changes in behaviour, and physiological stress.

This report is one part of a larger group project examining the tension between whale watching tourism and the disturbance associated with it— a tension the International Whaling Commission has tried to manage since first issuing whale-watching principles in 1996 *(International Whaling Commission, n.d.-a)*. My focus is the biological cost, the evidence for behavioural and physiological disturbance caused by vessels and their noise.

The broader social change question isn't really whether whale watching is "good" or "bad" for whales. It's more about how the industry can be managed so people can keep interacting with whales without imposing unacceptable biological costs on the animals being watched.

2. Public Discourse and Media

The public conversation around whale-watching disturbance takes several forms, each reaching different audiences. How disturbance is communicated shapes what the public understands as the problem.

One form is **viral incident coverage.** In May 2026, footage circulated of a commercial whale-watching vessel sitting close to orcas near the Gulf Islands, BC. CTV/CP24 reported on the incident as Fisheries and Oceans Canada opened an assessment *(CP24, 2026)*. The public can read the footage as visual evidence of harm, while the operator's response focused on compliance and the difficulty of judging distance through a telephoto lens. The regulator has to balance whale protection against the economic and cultural benefits of coastal tourism.

Another form is **regulatory discourse**. Canada's 2026 expansion of protective distance rules for endangered Southern Resident killer whales requires vessels to stay 1,000 metres away

(Fisheries and Oceans Canada, 2026). Unlike news coverage, this communication is procedural.

NGO advocacy communicates the issue differently again. Whale and Dolphin Conservation's "It's Rude to Intrude" campaign explains disturbance through relatable comparisons, such as being repeatedly woken up or having dinner interrupted, making disrupted feeding, resting, and nursing easier for a general audience to grasp *(Whale and Dolphin Conservation, 2019)*.

Science-adjacent communication frames the issue differently still, as something manageable through technology. A May 2026 report on AI systems used to listen for endangered orcas presents acoustic disturbance as a problem that can be monitored and reduced *(Mongabay, 2026)*.

Across these forms, the audience seems to shape what gets communicated: visual and emotional for viral coverage, procedural for regulators, relatable for NGOs, solutions-focused for technology. The most visible form of disturbance isn't necessarily the most biologically important one. A close approach can be filmed and shared immediately, while changes in respiration, stress hormones, or resting behaviour are largely invisible. That creates a bit of a communication gap between what people can see and what scientists actually measure.

3. Stakeholders

Biological disturbance from whale watching involves several stakeholders whose interests don't always align.

Whales and the marine ecosystem are an unusual kind of stakeholder: directly affected but unable to take part in public or policy discussion. Their interests— survival, undisturbed feeding, nursing, and breeding, end up having to be represented by scientists, conservation organisations, and regulators instead.

Whale-watching operators and the tourism industry depend on successful whale encounters for financial viability. They need access to whales while also needing clear, workable regulations. This doesn't necessarily mean operators are opposed to conservation, since the industry ultimately depends on healthy whale populations; but tourist expectations of close or reliable encounters can create pressure that contributes to disturbance.

Tourists and the general public are both consumers and a source of social pressure. They want close encounters for enjoyment, education, or connection with nature, while also expecting the experience to be ethical and environmentally responsible. Their expectations

influence operator behaviour, while media coverage and conservation campaigns influence what they consider acceptable.

Marine scientists measure behavioural, physiological, and population-level responses. Their role is to identify measurable effects and work out what the evidence can actually support.

Government and regulatory bodies translate scientific evidence into enforceable rules while weighing tourism, local economies, and safety, which is a genuine challenge, since scientific evidence carries uncertainty while regulation requires decision-making about distances, speeds, vessel numbers, and licensing.

NGOs and conservation organisations advocate for whales and translate scientific findings into public facing arguments, increasing pressure for stronger protection, though their advocacy naturally presents findings through a conservation lens.

These stakeholders overlap and are interconnected: research influences regulation, regulation affects operators, operators shape tourist experience, tourists drive demand, and NGOs and media shape public pressure on regulators. The International Whaling Commission frames whale-watching management in a similar way: as a multi-stakeholder process *(International Whaling Commission, n.d.-b)*.

4. Relevant Evidence

The strongest evidence for biological disturbance comes from studies connecting vessel presence or noise with measurable physiological or behavioural change.

Rolland et al. (2012) used the sharp, unplanned drop in shipping traffic following 11 September 2001 as a natural experiment, comparing underwater noise with stress-related faecal hormone metabolites in North Atlantic right whales before and after the drop. Lower low-frequency noise was associated with lower stress hormone levels. That's a useful setup because vessel presence and noise usually change together with other environmental factors, which normally makes them hard to pull apart.

Lemos et al. (2022) examined a similar relationship in gray whales off the Oregon coast, comparing vessel counts and ambient noise with faecal glucocorticoid concentrations across three field seasons. Vessel traffic the day before sample collection was significantly associated with stress hormone levels, even after controlling for sex, age, and reproductive state.

Sprogis, Videsen and Madsen (2020) probably offer the strongest direct connection to whale watching specifically. They played recorded vessel noise at different volumes to resting humpback mother-calf pairs off Western Australia. Louder playback reduced

resting time by around 30%, doubled respiration rate, and increased swim speed by over a third. This controlled experiment suggests that vessel noise itself can produce measurable responses, not just the presence of vessels.

Together, these studies cover three species, different regions, and both observational and experimental approaches. The evidence does have limitations, though:

- Much of it is correlational, so causation isn't always established
- Sample sizes for large-whale studies are limited
- Responses may differ between species, populations, and contexts
- Short-term behavioural or physiological changes don't automatically demonstrate long-term effects on reproductive success or population survival

Overall, the evidence seems to support the claim that vessels and their noise cause measurable disturbance in whales. Evidence linking these effects specifically to whale watching is strongest in controlled studies such as *Sprogis et al. (2020),* while broader vessel-noise research provides more general supporting context. It's still fairly unclear to what extent short-term effects translate into harm at the population level.

5. Discussion and Synthesis

Putting the discourse, stakeholders, and evidence together points to a central tension: the evidence is complex and uncertain, while public and regulatory decisions often need simple answers.

Viral footage can make a close approach look like the whole problem, while research shows acoustic disturbance and physiological stress can happen without being visually obvious at all. A minimum approach distance is a clear rule, but biological responses probably aren't determined by distance alone; vessel noise, speed, duration of exposure, number of vessels, and whale activity all seem to matter too.

Stakeholders also carry different priorities: operators need clear, workable rules; tourists want meaningful encounters; scientists need to account for uncertainty; regulators need enforceable thresholds; conservation organisations generally favour precaution wherever biological harm is possible. Research on the Australian whale-watching industry found environmental managers tended to prioritise species health and biological issues, while operators wanted clearer guidance on compliance, legislation, and rules *(Dimmock et al., 2014)*.

Social change probably isn't going to come from producing more scientific evidence alone. That evidence also has to be communicated effectively and translated into something stakeholders can actually understand and follow.

My position is that whale watching shouldn't necessarily be rejected, but should be managed around evidence-based limits on vessel proximity, noise, and interaction. Management should also be adaptive, given the uncertainty around long-term population effects, with rules monitored and revised as better evidence comes in. This recognises the social and economic value of whale watching while acknowledging that the industry depends, in the end, on healthy whale populations.

The social-change challenge, then, comes down to the industry's ability to adapt so that visiting whales doesn't end up threatening the biological conditions that make those experiences possible in the first place.

6. AI Use Acknowledgement

AI was used to help frame the references in the APA style, and cross check references and citations as well as check for broken links.
Chat link: https://share.gemini.google/HXXz03YEODoT

7. References

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